

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: CRISISBENCH | Context: Argentina — Buenos Aires Policy Maker Cohort (Disaster Reporting Reliability)

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotation provenance is minimally documented: the only directly attested annotation protocol is that AIDR data was labeled by domain experts **[Q111]**. For the remaining seven source datasets, no annotator demographics, recruitment methods, or inter-annotator agreement statistics are reported in the paper. Web research confirms CrisisNLP used crowdsourced workers with three-annotator agreement **[WEB-18]**, but no Argentine annotators or Argentine-norm calibration is documented anywhere in the provenance chain. Cross-dataset evaluation reveals a 14.3% F1 drop when training on CrisisLex and evaluating on CrisisNLP **[Q87]**, demonstrating that label conventions are not fully portable across event contexts — which raises substantial concern about portability to Argentine discourse. Dataset analysis empirically reveals concrete annotation noise: a Chilean cinema tweet labelled displaced_and_evacuations **[DATASET-D1]**, NGO mission descriptions labelled requests_or_needs **[DATASET-D37, D38]**, substantive typhoon warnings labelled not_informative **[DATASET-D28, D30]**, philosophical text labelled informative **[DATASET-D11]**, and Spanish-language disaster warnings systematically labelled not_informative **[DATASET-D9, D8]** — suggesting potential systematic under-recognition of Spanish-language disaster signals. Elicitation Q4 indicates Argentine professionals largely accept Global North label conventions, which softens but does not eliminate the concern.

ID	Evidence
Q4	“First, few efforts have been invested to develop standard datasets (specifically, train/dev/test splits) and benchmarks for the community to compare their results, models, and techniques.” (p.1)
Q87	“The cross dataset (i.e., train on CrisisLex and evaluate on CrisisNLP) results shows that there is a drop in performance. For example, there is 14.3% difference in F1 on CrisisNLP data using the CrisisLex model for the informativeness task.” (p.8)
Q111	“The AIDR data has been labeled by domain experts using AIDR system and has been labeled during different events.” (p.11)
D1	El trabajo de la mejor Daniela López que conozco en pantalla grande #matarauhombe #chile #cinema... http://t.co/asAx6KqNvD — Spanish tweet about Chilean film, labelled displaced/evacuations — apparent annotation error
D8	RT @meteomostoles: El tifón #BOPHA,de cat.4 y 947hPa en su centro,va camino de Filipinas.Lleva vientos sostenidos entre 2010 y 249km/h h ... — Spanish tweet about Philippines typhoon — meteorological content labelled not_humanitarian
D9	Ojo con el exceso de ayuda por los efectos del #TerremotoenChile porque puede producir mucho más obstáculos de lo que se cree. — Chilean Spanish warning about over-donation — labelled not_informative despite substantive crisis-relevant content
D11	when the ambushes of this century, hold closes even when life puts you in discomfiture, hold closes even that one of condescends your family, so God will facilitate you shortcoming — Philosophical/religious text labelled informative — no disaster content
D28	A tropical cyclone will affect my area. Winds of greater than 100 kph up to 185 kph may be expected in at least 18 hours. #TyphoonHagupit — Substantive typhoon warning labelled not_informative — clear label inconsistency
D30	Bangladesh:palazzo non era per fabbriche: Il progetto del Rana Plaza prevedeva solo 6 piani non 9 o 10 http://t.co/VG9qCqa3xX — Italian investigative tweet about Rana Plaza building code — labelled not_informative
D37	Equipped with a power generator and air conditioner, the tents can also house an emergency medical center. — Capability description mislabelled as requests_or_needs — annotation inconsistency
D38	AmeriCares solicits donations of medicines, medical supplies, and other relief materials from manufacturers, and delivers them quickly and reliably to indigenous health and welfare professionals in 137 countries around the world. — NGO capability description labelled requests_or_needs — mislabel
WEB-18	CrisisNLP used crowdsourced annotation with three-annotator agreement; demographics not published (https://aclanthology.org/L16-1259.pdf)

Strengths

- Argentine policy professionals draw on international (Global North) disaster communication standards as reference points [elicitation Q4], reducing expected systematic label disagreement for formal content
- AIDR subset has documented domain-expert annotation [Q111], providing a higher-confidence subset for evaluation

ID	Evidence
Q4	"First, few efforts have been invested to develop standard datasets (specifically, train/dev/test splits) and benchmarks for the community to compare their results, models, and techniques." (p.1)
Q111	"The AIDR data has been labeled by domain experts using AIDR system and has been labeled during different events." (p.11)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Partially — international (OCHA/ISCRAM-derived) label conventions used in the benchmark are recognized by Argentine emergency professionals [elicitation Q4], but Argentine institutional and journalistic norms for what constitutes a 'real signal' of disaster impact are not directly represented. Edge cases in political framing and informal-channel content are not aligned.

OC-2: Likely disagreement at margins: Argentine political satire of government crisis response and lunfardo-inflected commentary may be misclassified. Dataset analysis shows Spanish-language disaster warnings labelled not_informative [DATASET-D9, D8], suggesting systematic Spanish-content under-recognition that would propagate to Argentine deployment.

OC-3: Annotator demographics are not reported in the paper for seven of eight source datasets. Only AIDR is documented as domain-expert annotated [Q111]. CrisisNLP used crowdsourcing with three-annotator agreement [WEB-18] but demographics are not published. No Argentine annotators are documented.

OC-4: Re-annotation by an Argentine annotator pool would be advisable for any deployment-critical evaluation; this is user-acknowledged as a gap (elicitation flag).

OC-5: INSUFFICIENT DOCUMENTATION — paper does not describe aggregation methods or how minority annotator perspectives were handled across the eight constituent datasets.

OC-6: Documented label issues: (a) annotator demographics undocumented for 7/8 sources; (b) 14.3% cross-dataset F1 drop signals non-portable conventions [Q87]; (c) empirically observable annotation noise [DATASET-D1, D37, D38, D28, D33]; (d) Spanish-content label asymmetry [DATASET-D9, D8]. These violate convergent and external validity for Argentine deployment.

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Q87	"The cross dataset (i.e., train on CrisisLex and evaluate on CrisisNLP) results shows that there is a drop in performance. For example, there is 14.3% difference in F1 on CrisisNLP data using the CrisisLex model for the informativeness task." (p.8)
Q94	"Social media data is noisy and it often poses a challenge for labeling and training classifiers." (p.9)
Q111	"The AIDR data has been labeled by domain experts using AIDR system and has been labeled during different events." (p.11)

D1	El trabajo de la mejor Daniela López que conozco en pantalla grande #matarauhombe #chile #cinema... http://t.co/asAx6KqNvD — Spanish tweet about Chilean film, labelled displaced/evacuations — apparent annotation error
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D28	A tropical cyclone will affect my area. Winds of greater than 100 kph up to 185 kph may be expected in at least 18 hours. #TyphoonHagupit — Substantive typhoon warning labelled not_informative — clear label inconsistency
D33	Check out these award winning Long Island renovations: #longisland #suffolk https://t.co/wLnIbOENK9 — Completely unrelated home renovations tweet in other_relevant_information
D37	Equipped with a power generator and air conditioner, the tents can also house an emergency medical center. — Capability description mislabelled as requests_or_needs — annotation inconsistency
D38	AmeriCares solicits donations of medicines, medical supplies, and other relief materials from manufacturers, and delivers them quickly and reliably to indigenous health and welfare professionals in 137 countries around the world. — NGO capability description labelled requests_or_needs — mislabel
WEB-16	CrisisLex T6 used crowdsourced workers for relatedness labels; no Argentine annotators documented (https://crisislex.org/data-collections.html)
WEB-18	CrisisNLP used crowdsourced annotation with three-annotator agreement; demographics not published (https://aclanthology.org/L16-1259.pdf)

Information Gaps

- Annotator demographics, education levels, and recruitment for 7 of 8 source datasets
- Inter-annotator agreement statistics across source datasets
- Aggregation methods used to resolve disagreement across multi-annotator labels
- Whether the observed Spanish-language label asymmetry is systematic or sampling artifact

Requires Expert Verification

- Argentine policy/journalism expert re-annotation of a stratified Spanish-language sample to quantify expected label disagreement
- Validation of whether AIDR's domain-expert subset performs differently on Argentine-relevant content than crowdsourced subsets

Remediation

Gap 1: Undocumented annotator demographics for 7 of 8 source datasets; observed annotation noise and systematic Spanish-language label asymmetry

Recommendation: Conduct re-annotation of a stratified Spanish-language and Argentine-political-satire sample by an Argentine policy/journalism expert pool to quantify expected label disagreement. Restrict deployment-critical evaluation to AIDR domain-expert-annotated subset where possible. Publish a Datasheet documenting Argentine annotator demographics and IAA.